شماره دانشجویی:

نام و نام خانوادگی:

شماره صندلی: نام استاد:

دانشگاه صنعتی شریف

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شماره سوال:

$$P_{\theta,A}(\hat{x}) = P_Z(z)\left|\det\left(\frac{\partial f_\theta(z)}{\partial z}\right)\right|^{-1} = P_Z(g_\theta(\hat{x}))\left|\det\left(\frac{\partial g_\theta(\hat{x})}{\partial \hat{x}}\right)\right|$$ (الف

$$P_{\theta,A}(\hat{x}) = P_{\theta,A}\left(\begin{bmatrix} x \\ a \end{bmatrix}\right) \triangleq P_{\theta,A}(x,a) = P_\theta(x)\,P_A(a|x)$$ (ب

$$P_\theta(x) = \int_A da\, P_{\theta,A}(x,a)$$

$$P_\theta(x) = \int_A da\, P_{\theta,A}(x,a) = \int_A da\, q_\varphi(a|x)\frac{P_{\theta,A}(x,a)}{q_\varphi(a|x)} = \mathbb{E}_{a\sim q_\varphi(\cdot|x)}\left[\frac{P_{\theta,A}(x,a)}{q_\varphi(a|x)}\right]$$ (2.

$$\log P_\theta(x) = \log\left(\mathbb{E}_{a\sim q_\varphi(\cdot|x)}\left[\frac{P_{\theta,A}(x,a)}{q_\varphi(a|x)}\right]\right) \geq \mathbb{E}_{a\sim q_\varphi(\cdot|x)}\left[\log\left(\frac{P_{\theta,A}(\overbrace{x,a}^{\hat{x}})}{q_\varphi(a|x)}\right)\right]$$

$$= \mathbb{E}_{a\sim q_\varphi(\cdot|x)}\left[\log P_{\theta,A}(\hat{x}) - \log q_\varphi(a|x)\right]$$

$$= \mathbb{E}_{a\sim q_\varphi(\cdot|x)}\left[\log\left(P_Z(g_\theta(\hat{x}))\left|\det\left(\frac{\partial g_\theta(\hat{x})}{\partial \hat{x}}\right)\right|\right) - \log(q_\varphi(a|x))\right] \Rightarrow$$

$$\log P_\theta(x) \geq \text{ELBO}(x) = \mathbb{E}_{a\sim q_\varphi(\cdot|x)}\left[\log P_Z(g_\theta(\hat{x})) + \log\left|\det\left(\frac{\partial g_\theta(\hat{x})}{\partial \hat{x}}\right)\right| - \log q_\varphi(a|x)\right]$$

$$\mathcal{L}(\theta,\varphi;x) = -\mathbb{E}_{a\sim q_\varphi(\cdot|x)}\left[\log P_Z(g_\theta(\hat{x})) + \log\left|\det\left(\frac{\partial g_\theta(\hat{x})}{\partial \hat{x}}\right)\right| - \log q_\varphi(a|x)\right]$$ (ج

we need to find $\nabla_\theta \mathcal{L}$ and $\nabla_\varphi \mathcal{L}$.

to estimate loss, we need monte-carlo estimation $\rightsquigarrow$ samples from $q_\varphi(\cdot|x)$.

gradients need to pass through sampling for $\nabla_\varphi \mathcal{L}$ $\rightsquigarrow$ Reparametrization / Reinforce.

e.g. for gaussian q_φ: $a = \mu_\varphi(x) + \sigma_\varphi(x)\cdot\epsilon \quad \epsilon \sim \mathcal{N}(0, I)$

for generality, I assume a batch size of B.

Training process:

1. Sample B samples from $q_\varphi(\cdot|x)$: $a_i \sim q_\varphi(\cdot|x)$ "presumably has reparametrization"

2. construct: $\hat{x}_i = \begin{bmatrix} x \\ a_i \end{bmatrix}$

3. Forward pass: compute $z_i = g_\theta(\hat{x}_i)$ and $\log\left|\det\left(\overbrace{\frac{\partial g_\theta(\hat{x}_i)}{\partial \hat{x}}}^{J}\right)\right|$.

4. estimate loss:

$$L = \frac{1}{B}\sum_{i=1}^{B} -\log P_z(\underbrace{g_\theta(\hat{x}_i)}^{z_i}) - \overbrace{\log\left|\det\left(\frac{\partial g_\theta(\hat{x}_i)}{\partial \hat{x}}\right)\right|}^{\text{precomputed in fw pass}} + \log q_\varphi(a_i|x)$$

$$\underbrace{+ \overbrace{(\dots)}^{\text{signal}} \log q_\varphi(a_i|x)}$$

for Reinforce, extra if we have reparametrization.
this should not exist for reparam.

Reinforce: $\nabla_\varphi \mathbb{E}_{a \sim q_\varphi(\cdot|x)}[\ell(x,\theta,\varphi)] = \mathbb{E}_{a \sim q_\varphi(\cdot|x)}[\nabla_\varphi \ell(x,\theta,\varphi) + \ell(x,\theta,\varphi)\nabla_\varphi \log q_\varphi(a|x)]$

5. Compute backward pass: $\nabla_\theta L$, $\nabla_\varphi L$

6: update parameters: $\theta \leftarrow \theta - \eta_\theta \nabla_\theta L$

$$\varphi \leftarrow \varphi - \eta_\varphi \nabla_\varphi L$$

Reinforce correction is not needed. we have reparametrization.

Sampling / Inference:

1. $z \sim p_z(z)$

2. $\hat{x} = f_\theta(x) = \begin{bmatrix} x \\ a \end{bmatrix}$

3. discard a, return x as sample

Sampling mirrors standard NF sampling.

$$\log p_\theta(x) \geq \mathbb{E}_{a \sim q_\varphi(\cdot|x)} \left[\log p_z(g_\theta(\hat{x})) + \log \left| \det \left(\frac{\partial g_\theta(\hat{x})}{\partial \hat{x}} \right) \right| - \log q_\varphi(a|x) \right] = \text{ELBO}(x)$$

Estimation:

Similar to training process:

Simply estimate $\log p_\theta(x) \approx \text{ELBO}(x)$, and estimate $\text{ELBO}(x)$ using monte-carlo estimation.

Improvement:

1. larger sample size for monte-carlo estimation.

2. tighten $\log p_\theta(x) - \text{ELBO}(x)$ for this specific x, by maximizing $\text{ELBO}(x, \theta; \varphi)$ w.r.t. φ.